

BRYLE JAMES F. AGRA

INFORMATION TECHNOLOGY STUDENT

CONTACT INFORMATION

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Pangasinan, Philippines

PORTFOLIO

bryleagra.vercel.app

https://github.com/PancitGuisado

TECHNICAL SKILLS

- Full Stack Web Development
- Mobile & Cross-Platform Engineering
- RESTful API Architecture
- AI & Agentic Workflows
- Database Architecture & Modeling
- UI/UX Design Systems & Prototyping
- Interactive Canvas & Game Loop Engineering

LANGUAGES, TOOLS & FRAMEWORKS

Languages
TypeScript, JavaScript (ES6+), Python, PHP, SQL, HTML5, CSS3

Frontend & Mobile
React, Vite, Tailwind CSS, shadcn/ui, HTML5 Canvas API, Capacitor, Android APK, Responsive Design

Backend & APIs
Node.js, Express.js, PHP, RESTful APIs, Supabase Edge Functions

Databases & Cloud
PostgreSQL, MySQL, Supabase, Firebase, Database Design, Vercel

Developer Tools & Design
Git, GitHub, Figma, Postman, VS Code

AI & Agentic Systems
Claude, ChatGPT, Gemini, Antigravity, Claude Code, Codex

CORE WORK SKILLS

- Complex Problem Solving
- Computational & Algorithmic Thinking
- Clean Code & Maintainable Architecture
- Teamwork & Collaboration
- Adaptability & Technical Agility
- Project Management
- Continuous Learning & Trainability

PROFILE SUMMARY

I am a Full-Stack Developer passionate about transforming ideas into functional, high-performance digital products with clean code and intuitive interfaces.
With over 3 years of development experience, I specialize in building full-lifecycle applications across modern web and cross-platform ecosystems. My core stack includes React, TypeScript, Node.js, Express, PHP, PostgreSQL, MySQL, Supabase, and Tailwind CSS.

EDUCATION AND CERTIFICATIONS

BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

PHINMAUniversityofPangasinan | 2023–Expected2026/2027

- Major in Business Informatics
- Full Stack Web Developer and UI/UX Designer, EduBridge (2026)

RELEVANT EXPERIENCE

EduBridge AI-Assisted Educational Platform — Lead Full-Stack Developer & UI/UX

CapstoneProject | 2025–2026

- Designed high-fidelity UI/UX design systems, user journeys, and responsive student/instructor dashboards in Figma, translating them into pixel-perfect React components.
- Engineered full-stack web architectures, REST APIs, and biometric face verification with OpenCV/DeepFace for fraud prevention.

The Signal — Lead Game Developer & Systems Engineer

Independent Cross-Platform Development | 2025–2026

- Engineered an interactive 2D game loop, collision detection systems, and wave-scaling mechanics with TypeScript and HTML5 Canvas running at a steady 60 FPS.
- Packaged and distributed cross-platform builds to Android using Capacitor, optimizing HUD interfaces, touch controls, and memory performance.

PROJECTS

EduBridge — AI-Assisted Educational Platform

CapstoneProject | Lingayen LGU | 2026

TypeScript | React | Node.js | Express.js | Supabase | PostgreSQL | AI Integration | Tailwind CSS

- Engineered an intelligent learning management platform combining structured curricula with artificial intelligence to optimize student study habits and teacher workflows.
- Implemented role-based authentication and secure database schemas in Supabase/PostgreSQL to partition administrative controls, assignment distribution, and student analytics.
- Built an interactive progress dashboard with real-time feedback, automated quizzes, and responsive UI components for cross-device accessibility.

The Signal — Cross-Platform 2D Tower Defense Survival Game

FeaturedProject | Game Dev & Mobile Systems | 2025–2026

TypeScript | React | HTML5 Canvas | Capacitor | Android APK | Tailwind CSS | Vite

- Developed a 2D pixel-art tower defense survival game featuring custom game loops, grid-based building mechanics, enemy pathfinding, and dynamic resource management.
- Packaged the web codebase into a standalone Android APK via Capacitor, incorporating mobile-optimized HUD modals, settings menus, and touch input controls.
- Optimized frame rendering, boss-exclusive item drop mechanics, and state persistence, ensuring smooth 60 FPS gameplay across desktop browsers and mobile devices.

AIPixelCraft — AI-Powered Pixel Art & Sprite Editor

FeaturedProject | Generative AI & Web Tools | 2025–2026

React | TypeScript | Generative AI | HTML5 Canvas API | Tailwind CSS | Vite

- Architected a browser-based creative suite that leverages artificial intelligence to generate, customize, and export game-ready 2D pixel art and sprite sheets.
- Integrated AI generation pipelines with custom grid-conforming prompts, allowing indie creators to rapidly prototype consistent game assets.
- Built a high-performance Canvas editor with features for layer editing, palette selection, real-time preview, and instant client-side asset export.

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INFORMATION TECHNOLOGY STUDENT • FEATURED PROJECTS & PORTFOLIO

OFFICIAL ACADEMIC CAPSTONE PROJECT 2026

EduBridge — AI-Assisted Educational & Classroom Management Platform

Role: Lead Full-Stack Web Developer & UI/UX Designer

A comprehensive educational platform integrating artificial intelligence with structured curricula to personalize student learning and streamline administrative workflows for educators, featuring automated grading assistance, progress tracking, and interactive study modules.

STACK: React • TypeScript • Node.js • Express.js • Supabase • PostgreSQL • OpenAI / Gemini API • Tailwind CSS • REST APIs

- ▸ AI-Powered Learning Assistant: Generates context-aware study guides, custom quizzes, and instant feedback tailored to individual student performance metrics.
- ▸ Role-Based Classroom Architecture: Built dedicated portals for students and instructors with secure authentication, gradebooks, and assignment distribution.
- ▸ Real-Time Progress Analytics: Delivers interactive data visualizations and mastery tracking, enabling educators to identify learning gaps and intervene early.

FEATURED AI PLATFORM 2026

Ai PixelCraft — AI-Powered Pixel Art & Sprite Generation Suite

Role: Lead Full-Stack Developer & Graphics Engineer

An advanced browser-based creative suite that leverages artificial intelligence to generate, customize, and slice game-ready 2D pixel art assets and dynamic sprite sheets for indie game developers.

STACK: React • TypeScript • HTML5 Canvas API • Generative AI Models • Tailwind CSS • Vite • Lucide Icons

- ▸ AI Analytics Engine: Evaluates transactional data using LLMs to forecast eAI Prompt-to-Sprite Engine: Integrates generative models tuned for pixel-perfect sprite
- ▸ Low-Latency Canvas Editor: Built an in-browser graphics manipulation tool supporting layered drawing, palette swapping, zoom/pan navigation, and sprite-sheet slicing.

AI PRODUCTIVITY PLATFORM 2026

Resumaker — AI-Powered Resume Builder & Document

Role: Full-Stack Developer & Product Designer

A streamlined web application that enables job seekers to construct ATS-optimized, professionally styled resumes effortlessly through AI-driven content generation, smart formatting, and real-time live preview.

STACK: React • TypeScript • Node.js • OpenAI API • Tailwind CSS • JsPDF

- ▸ Smart Content Recommendations: Evaluates target job roles and dynamically suggests impactful bullet points, relevant skills, and polished phrasing using LLMs.
- ▸ Dynamic Layout & PDF Export: Real-time WYSIWYG document section toggling, automated layout reflow, and instant vector PDF generation.

CROSS-PLATFORM (WEB & MOBILE) 2026

The Signal — 2D Cross-Platform Tower Defense Survival Game

Role: Lead Game Developer & Systems Engineer

An original 2D pixel-art top-down sideview tower defense survival game where players build defenses, manage resources, and survive escalating waves of alien incursions across desktop web browsers and native Android devices.

STACK: TypeScript • React • HTML5 Canvas API • Capacitor • Android (APK Build) • Tailwind CSS • Vite • Web Audio API

- ▸ Custom 2D Engine & Game Loops: Engineered collision detection, pathfinding algorithms, projectile physics, and boss-exclusive drops running at a steady 60 FPS.
- ▸ Native Mobile Deployment: Packaged the core application into a standalone Android APK via Capacitor, implementing mobile-optimized touch controls and HUD modals.
- ▸ Progression & Resource Architecture: Designed an upgrade economy featuring turret tiers, alien power core harvesting, wave-scaling algorithms, and persistent saves.

MOBILE PRODUCTIVITY 2024

EZ MART — Modern E-Commerce & Retail Management Platform

Role: UI/UX Designer

A specialized native Android mobile retail storefront and inventory platform providing seamless product discovery, secure checkout workflows, and automated order management.

STACK: Kotlin • Android Studio • SQLite • Material Design 3

- ▸ Intuitive Catalog & Cart Engine: Stateful client-side shopping cart with dynamic category filtering, search indexing, and real-time inventory availability checks.
- ▸ Order Management & Backend APIs: RESTful API services supporting secure checkout processing, customer authentication, and administrative inventory updates.

NEWS & MEDIA WEB 2023 — 2024

Students' Herald — Digital University Publication

Role: Frontend Developer & UI Contributor

The official web news and publication platform of PHINMA University of Pangasinan delivering student journalism, campus announcements, and editorial articles.

STACK: HTML5 • CSS3 • JavaScript • PHP • MySQL • Responsive UI

- ▸ Responsive Editorial Layout: Engineered mobile-first reading interfaces with fluid typography, article tagging, and fast asset loading.
- ▸ Editorial Workflow: Built clean content publishing modules with category filtering and cross-browser accessibility compliance.